

Pratham Kairamkonda

📍 Hyderabad, India

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Backend and data-focused AI developer with two internships and hands-on projects spanning enterprise pharma software and SQL/Django systems, local LLM pipelines, and multimodal AI – research to deployment, built and self-hosted from scratch.

EXPERIENCE

ProvenTech (an NTT Data Company)

01/2026 – 04/2026

Trainee Consultant (Intern)

Hyderabad, India

- Resolved **backend bugs across Django REST API and Microservices modules** in a live enterprise LIMS serving Cipla, Lupin, Dr. Reddy's, and BioCon, sustaining stability of critical manufacturing integrations.
- Authored **10+ optimized SQL queries** cutting retrieval time by **~30%** on live pharma reporting pipelines; cleaned duplicate records across SQL and MongoDB, improving reporting accuracy.

Swecha

07/2024 – 10/2024

Machine Learning Intern

Hyderabad, India

- Built a Computer Vision **Telugu OCR pipeline** (Python, Tesseract, OpenCV) hitting **71% character accuracy** and 30% faster throughput; digitized 10,000+ archival documents.
- Co-developed **Chandamama Kathalu**, a fine-tuned Telugu stories SLM expanding digital accessibility on Telugu language data.

PROJECTS

Project Sanjaya | Python, FastAPI, React, Llama-cpp, VLMs

- Built a Multi-model AI surveillance engine using **VLM and Computer Vision** that converts live video streams into queryable semantic text logs, eliminating long-term video storage while enabling vector database backed retrieval **designed to support across years of activity**.
- Exposed a real-time RAG interface over REST API (FastAPI); security summaries delivered on demand via Telegram.

VocalNote – AI PDF-to-Podcast Converter | Python, FastAPI, React, Kokoro TTS, Llama-cpp

- End-to-end PDF-to-audio pipeline using local LLM inference, Role based and One Shot Prompting and TTS Engine; optimized for consumer hardware with **real-time generation and no GPU required**.
- Officially recommended by the University IT Department**; adopted by **200+ students** within two weeks. Fully offline, zero-cloud LLM inference tool.

Telugu-OCR – Document Digitization Pipeline | Python, FastAPI, React, Tesseract, OpenCV

- Rebuilt a batch OCR pipeline into a bounded, production-grade extraction service: page-count discovery before rendering, hard page/upload caps, and per-page timeouts with graceful degradation instead of pipeline failure.
- Fixed a **thread-oversubscription bug** causing N^2 thread contention under page-level parallelism – the single highest-impact throughput fix in the pipeline; replaced global-mean binarization with adaptive Gaussian thresholding for better accuracy on unevenly lit scans.

Sutravak – Long-Form Audio Generation Pipeline | Python, FastAPI, Chatterbox TTS

- Built a sentence-boundary-aware chunking stage feeding a single-worker serialized job queue, preventing GPU/MPS resource exhaustion under concurrent requests.
- Precomputed and persisted waveform-peak data per job so the frontend renders instantly instead of decoding full audio client-side – a server-side precompute pattern keeping downstream consumers fast.

HOMELAB & INFRASTRUCTURE

Self-Hosted Infrastructure | *Docker, Nginx, Cloudflare Tunnel, Raspberry Pi*

- Designed and operate a self-hosted homelab, migrated from a repurposed laptop to a **Raspberry Pi 4 Model B (8GB RAM)**, running every service containerized under Docker on constrained CPU-only hardware.
- Exposed all services externally through a **Cloudflare Tunnel** with an Nginx reverse proxy, avoiding port-forwarding and static IP requirements, on a zero-paid-infrastructure setup (free domain via DigitalPlat Domains).
- Self-hosted and maintenance of **Media Server** (movies, shows and music), **self-hosted photo/video** (backup with ML-based search, self hosted Google Photos), **Navidrome** (Subsonic-compatible music streaming) hosting FLAC audios, **n8n** (workflow automation), and **Uptime Kuma** (service monitoring).
- Manage resource contention across concurrently running containerized services on a single low-power board; planning next steps around Migrating services to Docker Compose and centralized secrets management.

Narad – Personal AI Agent | *Python, Telegram Bot API, LLM Tool-Calling, Obsidian, Agent Reach*

- Built a 24/7 personal automation agent running on the homelab Pi with tool-calling access to the host system, driven via Telegram chat as the interface.
- Automates recurring dev/media workflows: downloading and converting media (e.g. YouTube video/audio extraction), moving files into the self-hosted media server, and retrieving photos/files from the homelab on demand – reducing manual SSH/file-transfer steps.

SKILLS

Languages : Python, Java, SQL, MongoDB, Bash/PowerShell

Frameworks & Web : Django, FastAPI, Streamlit, Flask, WebSockets, RestAPI, Microservices

AI/ML & Data : OpenCV, LangChain, LangGraph, RAG, LLM Quantization, Vector Databases, Fine-Tuning, Scikit-learn, Pandas/NumPy, MCP, Computer Vision, NLP, Prompt Engineering, Multimodal AI, Generative AI, vLLM

Tools & Infra : Git, GitHub, Docker, Docker Compose, GitHub Actions, CI/CD, Nginx, Cloudflare Tunnel, Raspberry Pi, Agile/Scrum, SQLite, pytest

EDUCATION

Anurag University

B.Tech in Information Technology | CGPA: 7.6/10

2022 – 2026

Hyderabad, India

Certifications

Wipro TalentNext – Java Full Stack | Cisco CCNA 1–3 | Python Essentials 1 & 2 | Google ML Crash Course